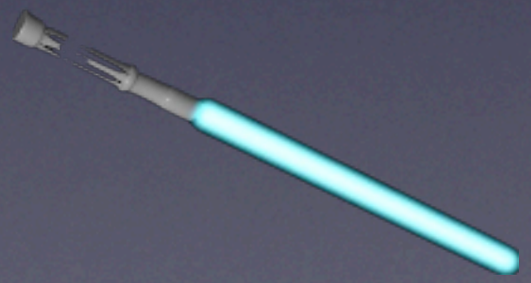
The War To Build a Quantum Computer

Dave Bacon
University of Washington







D-Wave joke deleted by Canadian border control



ENIAC, 1946

- 17,468 vacuum tubes
- 7,200 crystal diodes
- 1,500 relays
- 70,000 resistors
- 10,000 capacitors
- ~5,000,000 hand-soldered joints

~ one vacuum tube failure / day
(equivalent to ~ one transistor failure /second today)

Alternative History

Imagine: no transistors, no integrated circuits, just unreliable components...



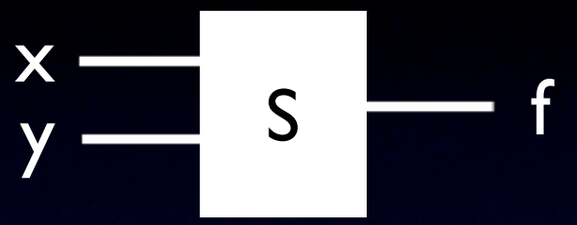




Could we build a computer out of unreliable components?

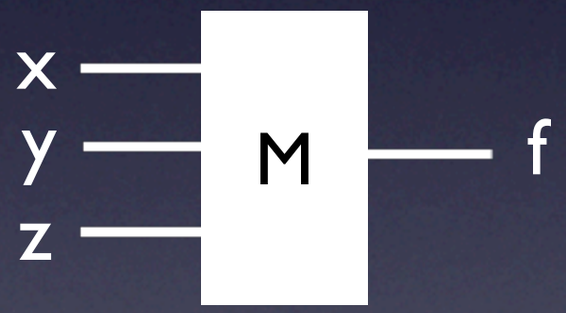
J. von Neumann, "Probabilistic Logics and the Synthesis of Reliable Organism from Unreliable Components" 1956

Von Neumann's Model



NAND gate

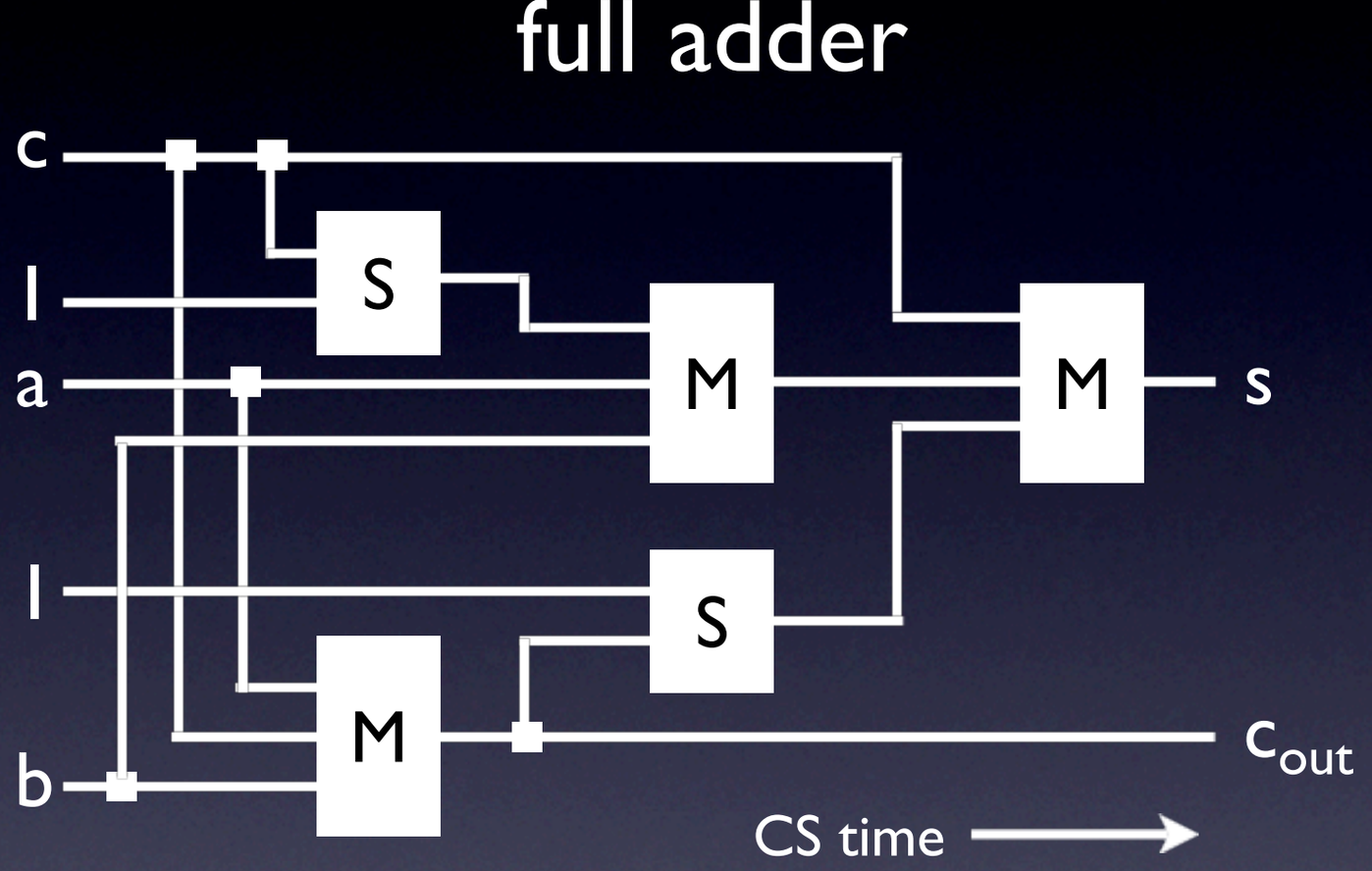
x	y	f
0	0	1
0	1	1
1	0	1
1	1	0



majority gate

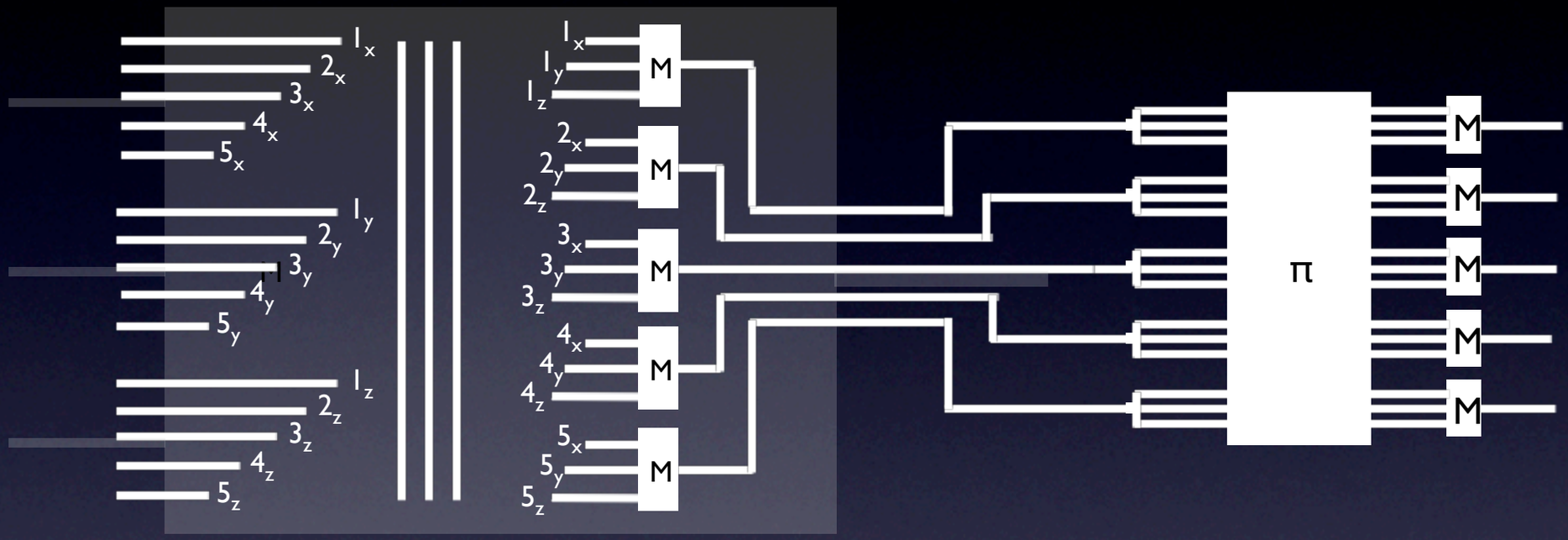
x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

full adder



suppose each gate fails (outputs wrong answer) *independently* and with probability p

Von Neumann's Solution

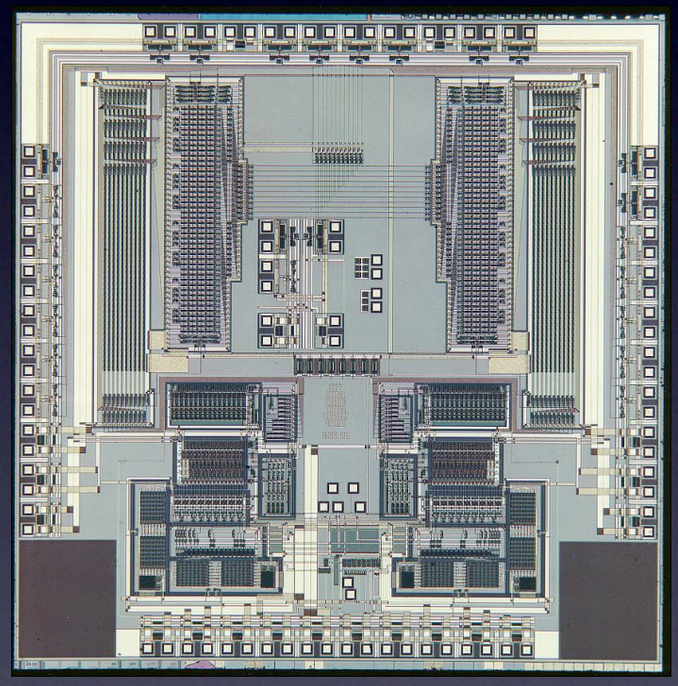


gate error correction

Theorem (von Neumann): A circuit of g gates can be made to fail with probability ε using $O(g \log (g/\varepsilon))$ gates which fail with probability p, assuming $p<p_{th}$ (vN estimated $p_{th} \sim 1\%$)

We Don't Need Your Stickin' Engineering







There are distinct *physical* reasons why robust classical computation is possible

Fault-Tolerant Quantum Computing

Consider quantum computing where all quantum circuit components can fail:



state preparation



gates

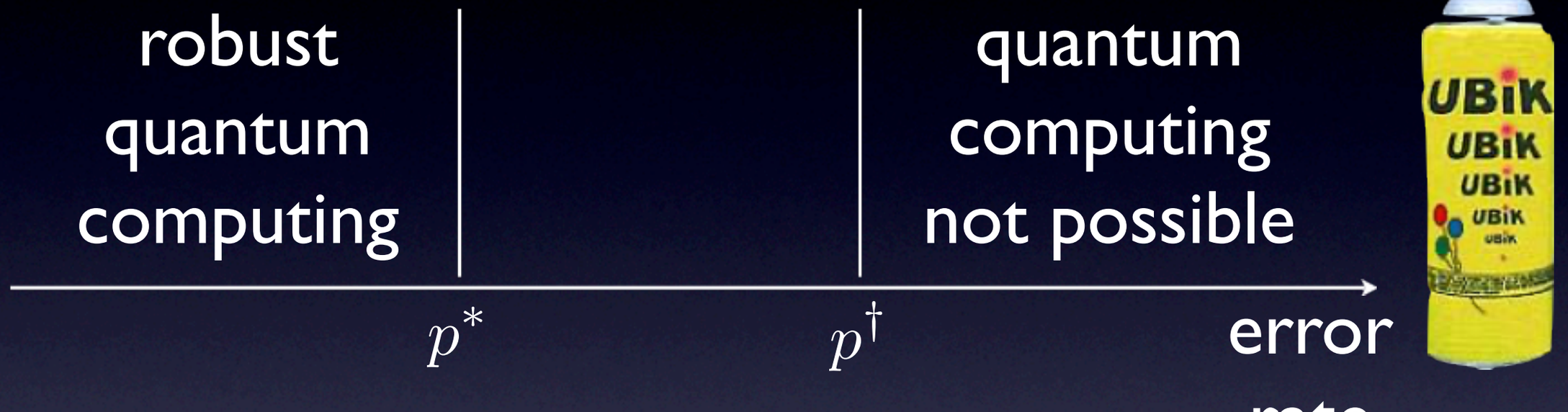


wires



measurements

The Quantum Threshold Theorem



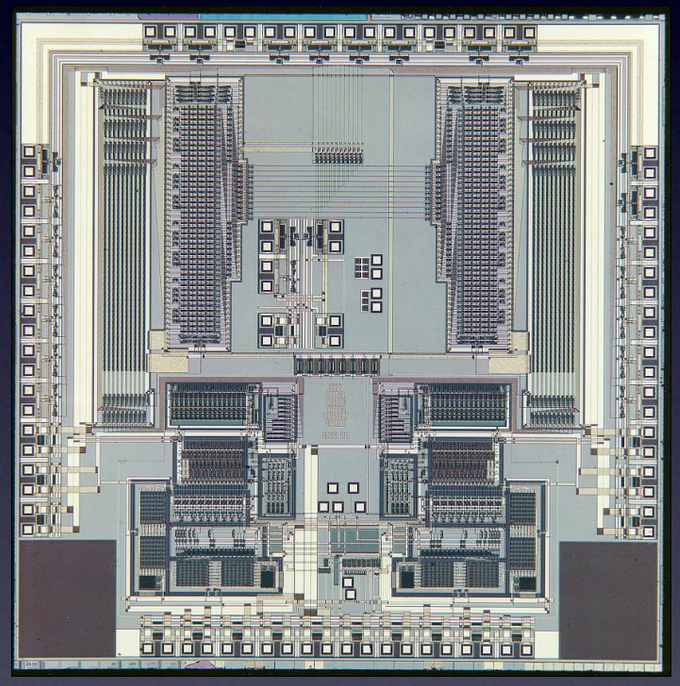


Theorem: If decoherence and lack of control are below a threshold p* (relative to coherent control), then a quantum computation of size g can be carried out with error probability ε using $O(g \log^c(g/\varepsilon))$ extra qubits

Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Kitaev, Gottesman, Preskill (1996)

But....







Kitaev's Idea

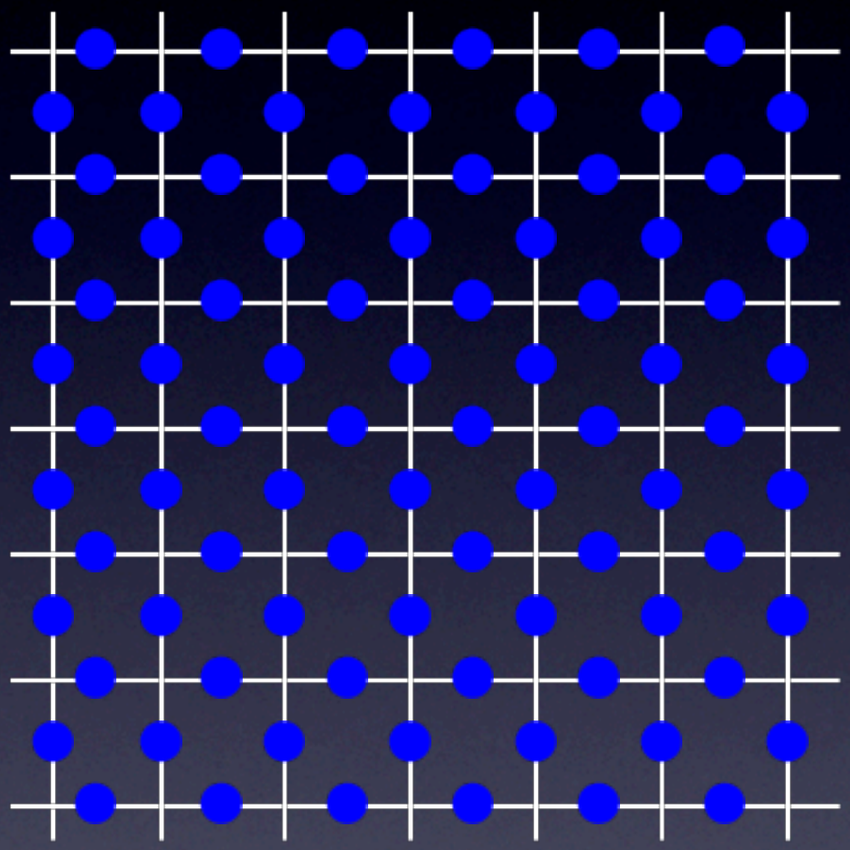


Alexei Kitaev
(Caltech)

"Magnetism arise from spins of individual atoms. Each spin is quite sensitive to thermal fluctuations. But the spins interact with each other and tend to be oriented in the same direction. If some spin flips to the opposite direction, the interaction forces it to flip back to the direction of other spins. This process is quite similar to the standard error correction procedure for the repetition code. We may say that errors are being corrected at the physical level. Can we propose something similar in the quantum case? Yes, but it is not so simple."

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)

Kitaev's Toric System



periodic boundary conditions (torus)

$\bullet = Z \quad \bullet = X$

plaquette operator



$$S_p = \prod_{e \in \delta P} Z_e$$

vertex operator



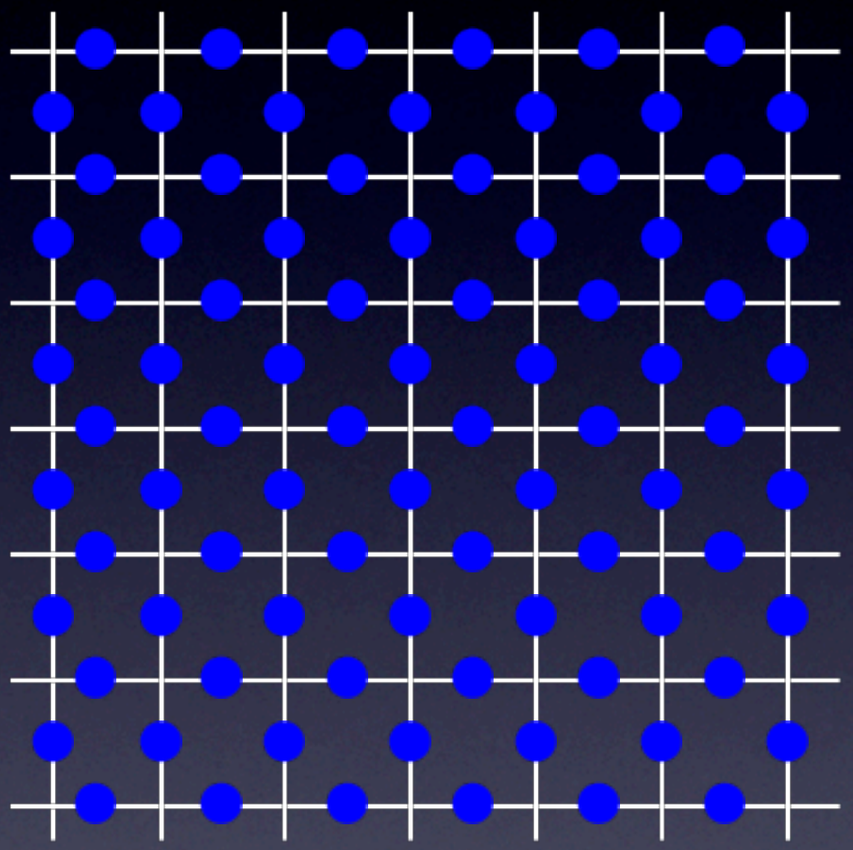
$$S_v = \prod_{e | v \in e} X_e$$

$$[S_p, S_v] = 0 \qquad S_p^2 = S_v^2 = I$$

Kitaev Hamiltonian:

$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

Kitaev's Toric System



Kitaev Hamiltonian:

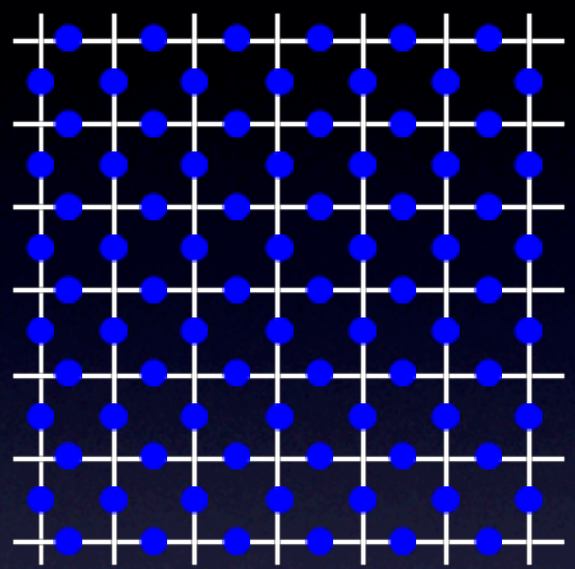
$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$



ground state is a quantum error correcting code

$2L^2$ qubits, encodes 2 qubits, can correct L/2 qubit errors

Kitaev's Toric Protection



Kitaev Hamiltonian:

$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

$$2\Delta$$

Single qubit random perturbations $V = \lambda \sum_e P_e$

$$\delta E = \frac{\lambda^L}{\Delta^{L-1}}$$

Exponential suppression of level splitting due to external (local) perturbations

Topological Quantum Computing



non-abelian Anyons can be used to build a universal quantum computer

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



quantum computing using representations of braid group

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)
Freedman, "Quantum computation and the localization of modular functors" (2001)

The War

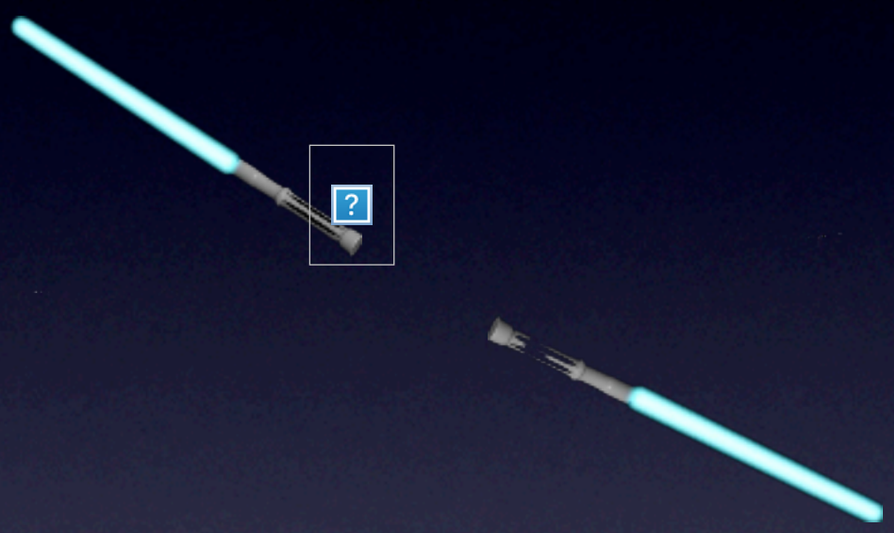
The Engineers

- Worship at the alter of the threshold theorem
- Considerations
  - Maximize threshold
  - System which can be “scaled up” while being under the threshold
  - Focus: build single and double qubits as robustly as possible...then scale to quantum error correction

The Physicists

- Use physics to protect quantum information
- Considerations
  - Must engineer system with many-body interactions
  - Theory needs to minimize complexity and suitability of such systems
  - Focus: many dirty interacting qubits

Whose Winning?



- Majority of experimental quantum computing research
- Short term hurdle: break even for a quantum error corrected system
- Long term: scaling, heat, size, speed

Whose Winning?



- Smaller effort

- Short term hurdle: observation of non-abelian anyons

- Long term: non-abelian anyons in universal quantum system, fault-tolerance of non-abelian anyon manipulations

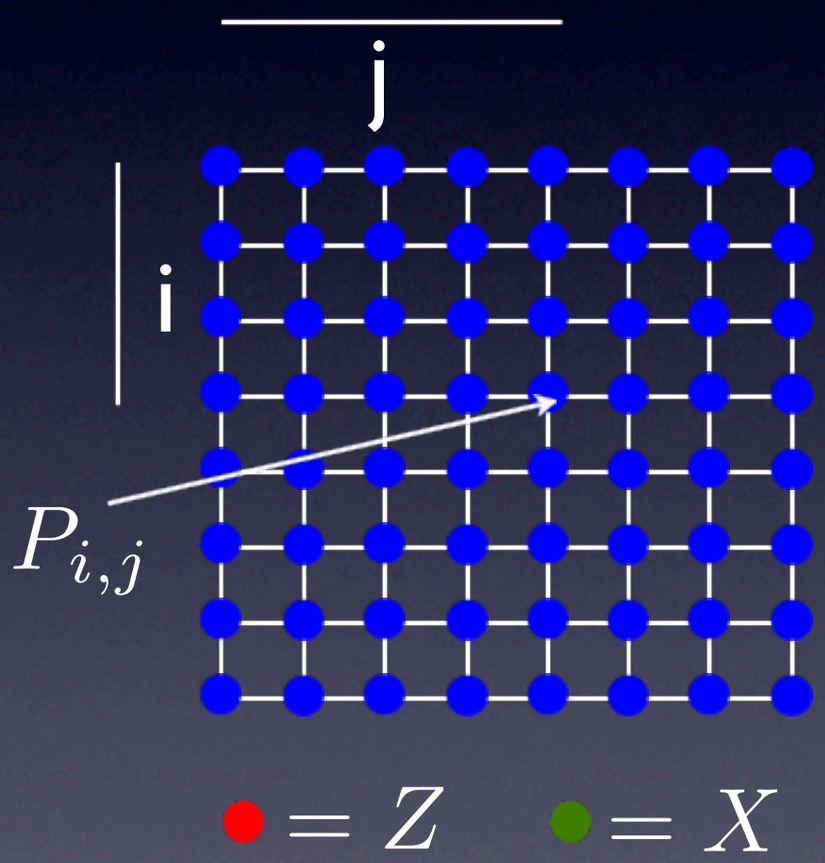
Peacemaker

Can we use ideas from the physicist side to help the computer engineering side?

Observation: physicists care an awful lot about the spatial locality of their models

Doubling the Toric Code

L x L (L even) square lattice



z type

$$S^z_{i,j} = Z_{i,j} Z_{i+1,j} Z_{i,j+1} Z_{i+1,j+1}$$

x type

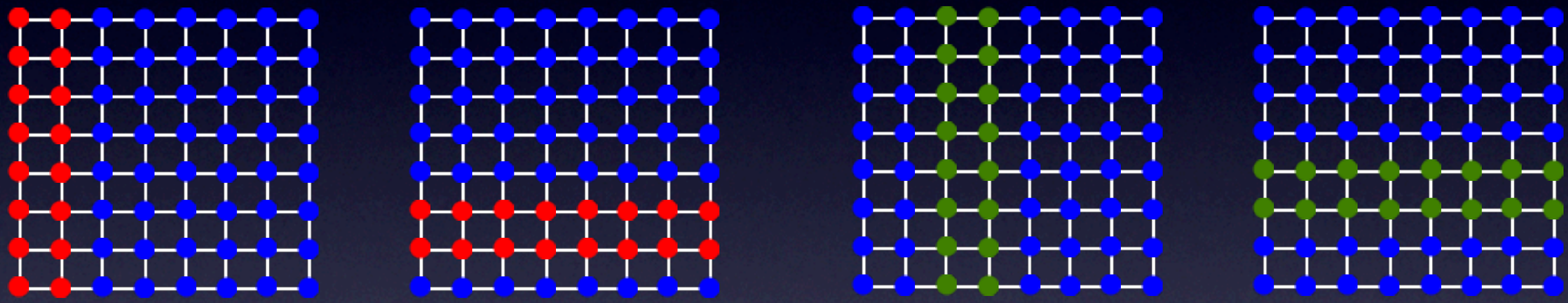
$$S^x_{i,j} = X_{i,j} X_{i+1,j} X_{i,j+1} X_{i+1,j+1}$$

Consider the group generated by these operators:

$$\mathcal{T} = \langle \quad , \quad \rangle$$

Doubled Toric Code

Abelian invariant subgroup:



$$C_j^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i+1,j} \quad C_i^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i,j+1} \quad C_j^X = \prod_{i=1}^{L} X_{i,j} X_{i+1,j} \quad C_i^X = \prod_{i=1}^{L} X_{i,j} X_{i,j+1}$$

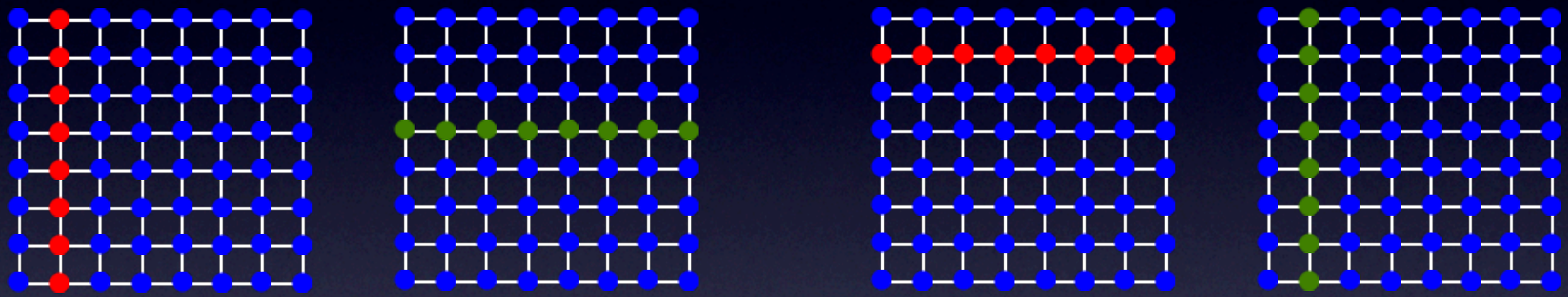
$(L-1)^2 - 1$ generators $(L-1)^2 - 1$ generators

$\bullet = Z \quad \bullet = X$

Stabilizer group

Doubled Toric Code

Normalizer of stabilizer group: $\mathcal{T}$ and two logical qubits



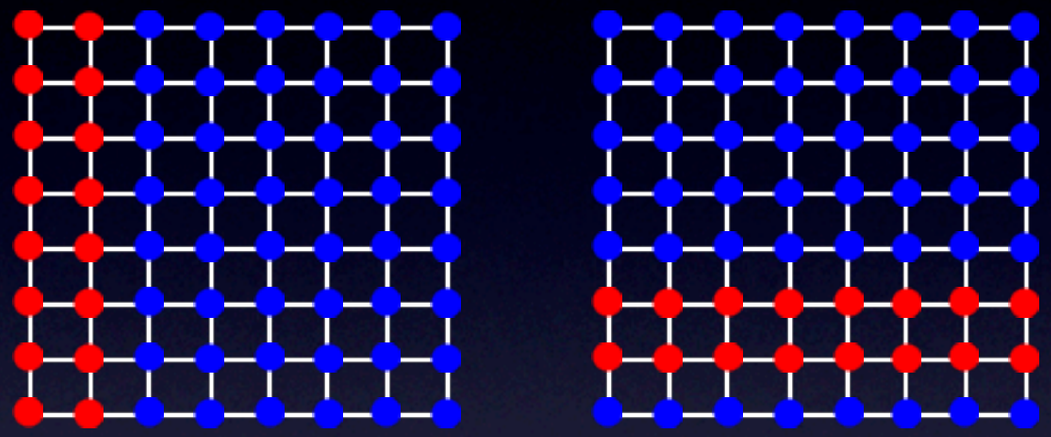
Z_1 X_1 Z_2 X_2

[[L^2,2,L^2-2,L/2]] subsystem code

● $= Z$ ● $= X$

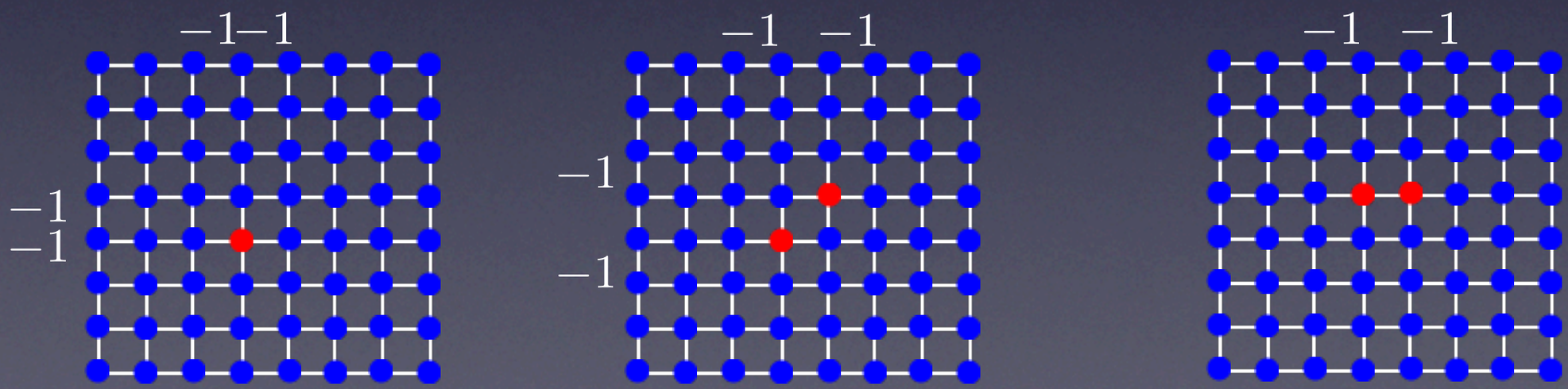
Doubled Toric Code

Error correcting procedure:



treat as redundancy code on boundary

$$C_j^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i+1,j} \qquad C_i^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i,j+1}$$



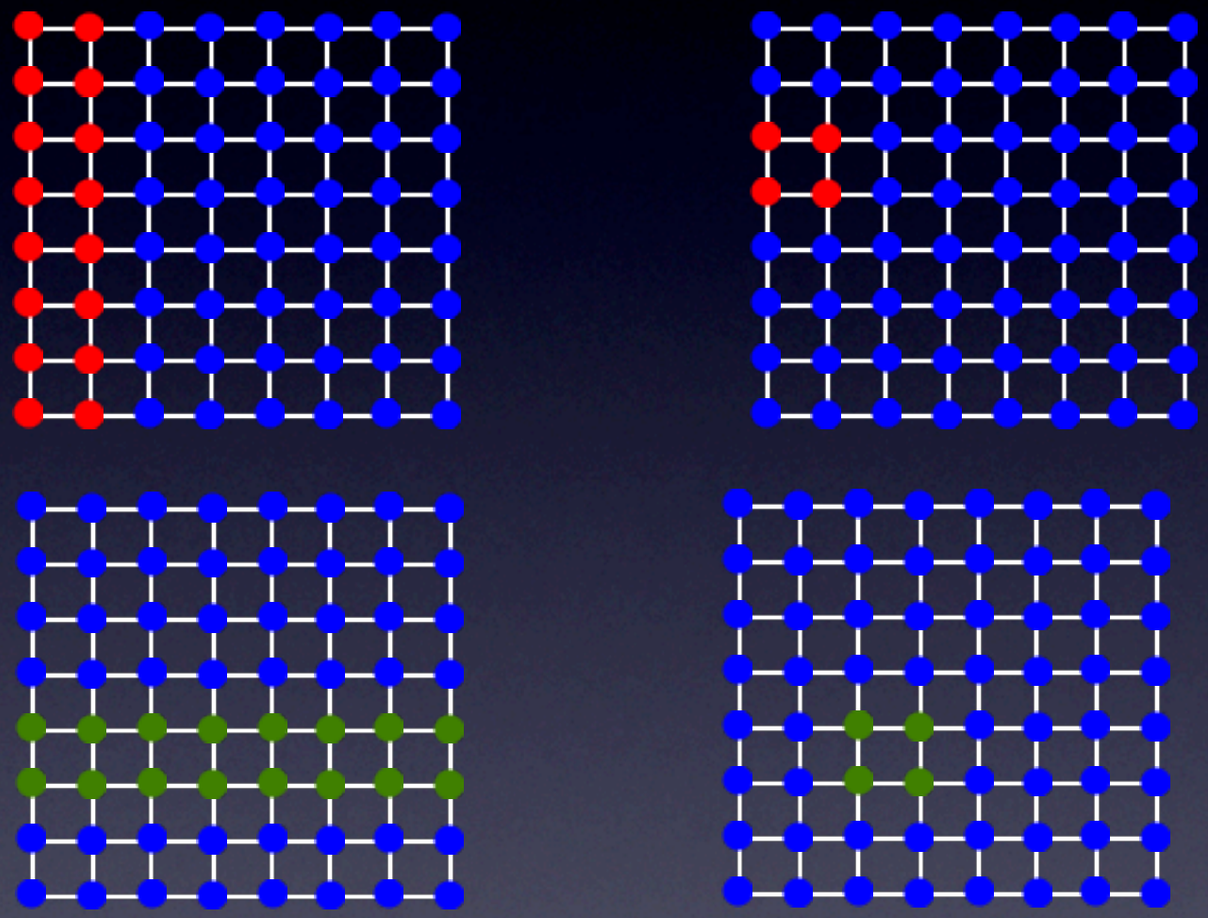
● $= Z$ ● $= X$

matched errors

unmatched

Doubled Toric Code

Fault tolerance and locality

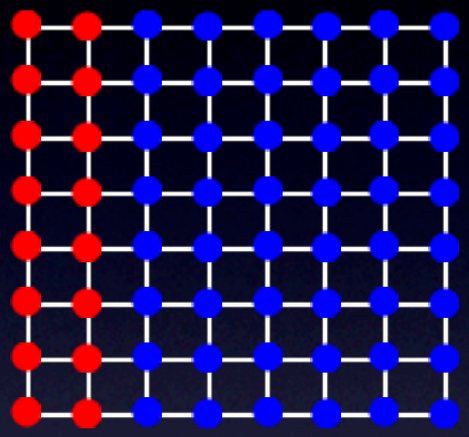


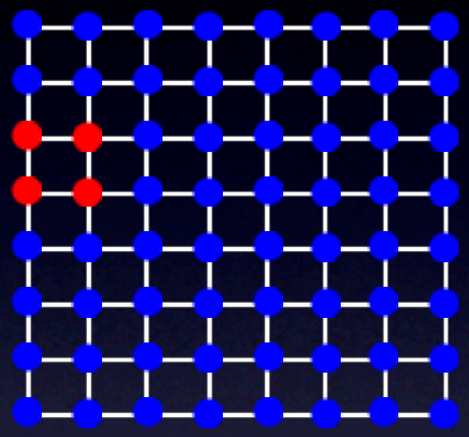
Even though they don't commute we can measure each four qubit operator separately then send info to boundary to correct

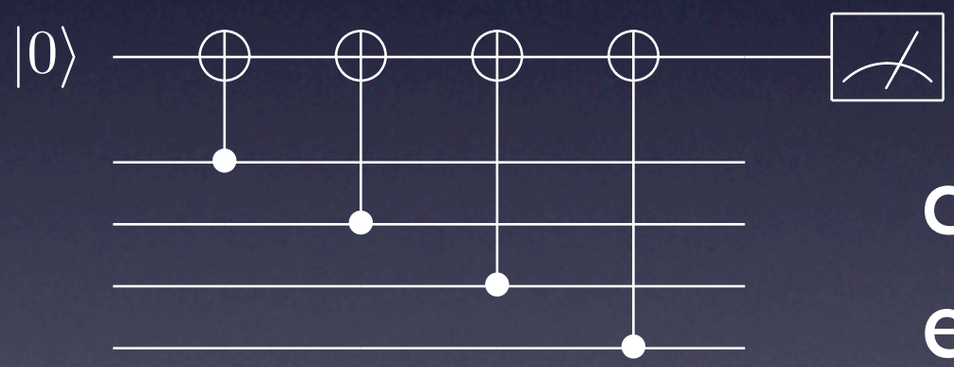
$\bullet = Z \quad \bullet = X$

Doubled Toric Code

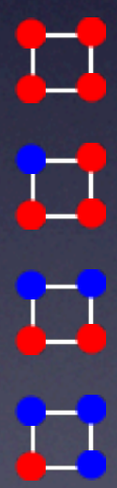
Fault tolerance and locality







can spread Z error to code



gauge

gauge + 1 err

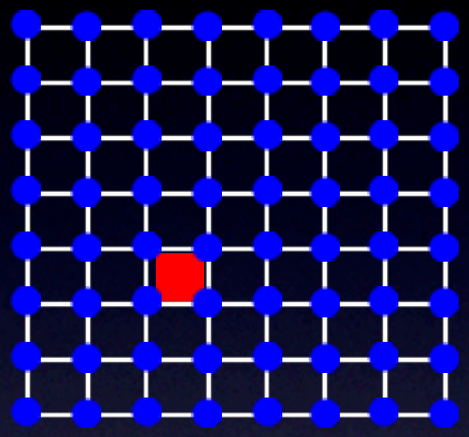
2 err

1 err

If distance of code >5, then 2 errs not a problem

$\bullet = Z \quad \bullet = X$

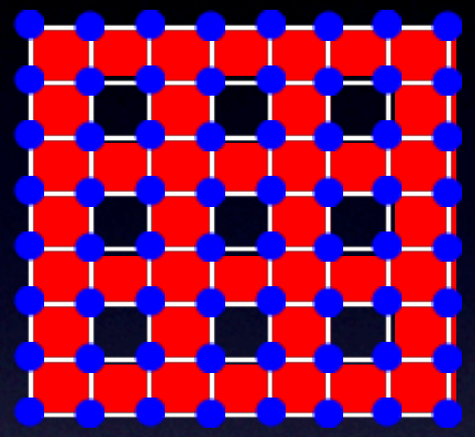
Puncturing the DTC



remove one

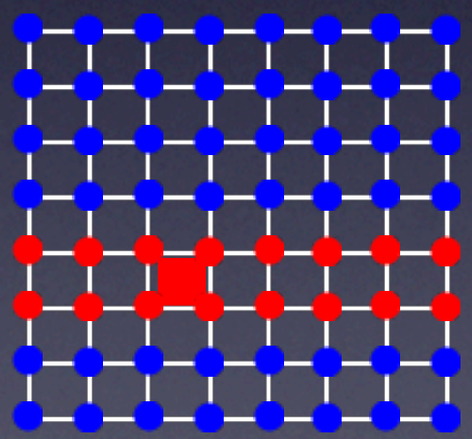


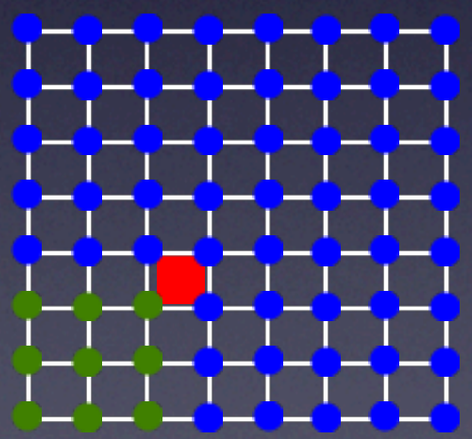
z type



region of allowed removal

Z_L



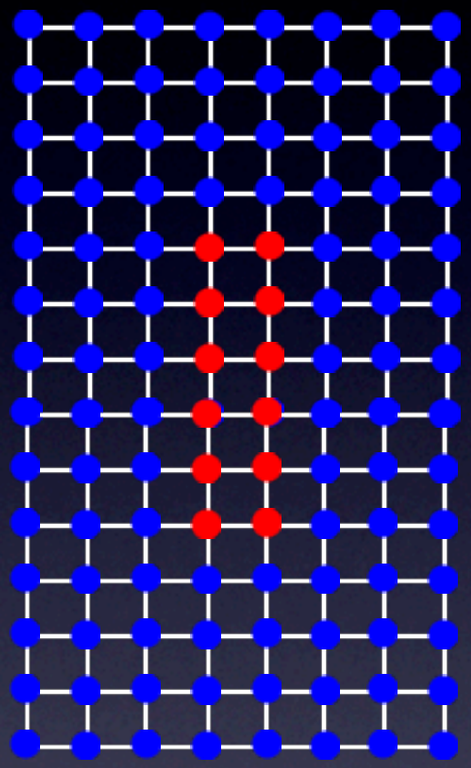


X_L

produces one encoded qubit

$\bullet = Z \quad \bullet = X$

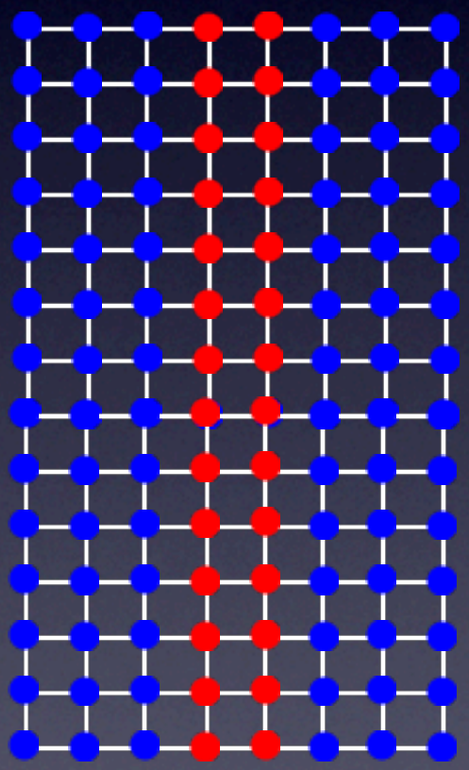
Puncturing the DTC



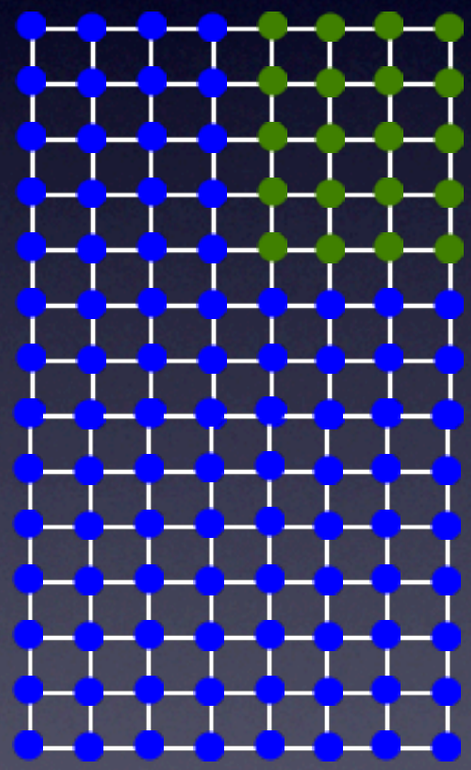
remove > 1?



z type



Z_L

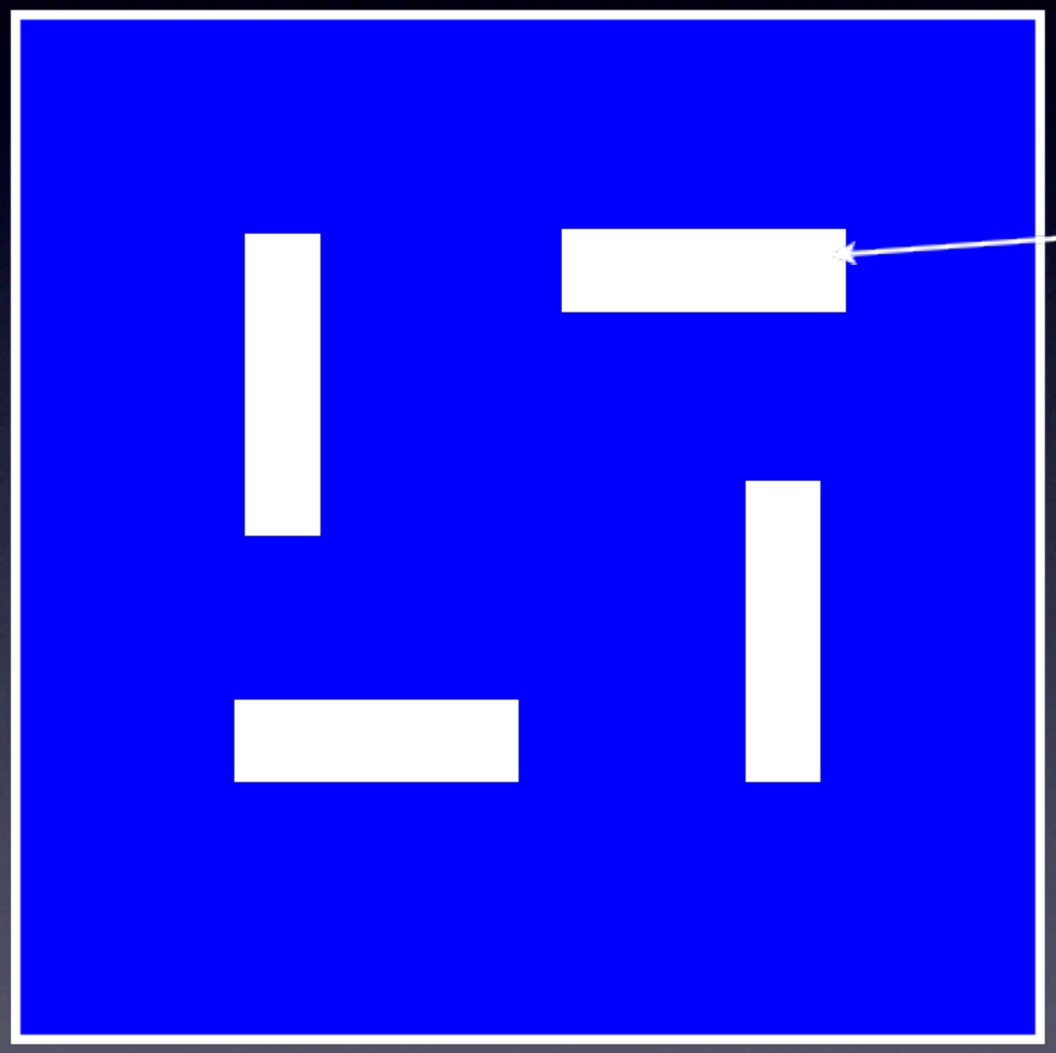


X_L

weight of smallest Z error (mod gauge) = length of defect

Local Architecture?

Build a nearly local quantum error correcting architecture

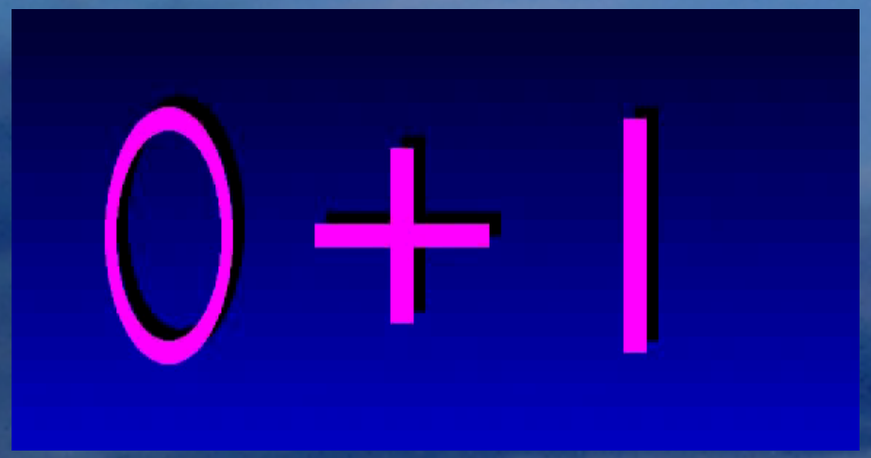


Benefit over toric surface code:
error correction at boundary

Fin

Fault-tolerant quantum computing:
physics or computer engineer?

Locally interacting quantum systems lead to codes with good spatial locality (doubled Kitaev code)



Quantum Computing Theory Group

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Alice Neels (CSE)

Masters Students
Roger Wolfson
(Physics)

Undergraduates
Elizabeth Muhm (CSE/Math)
Alper Sarikaya (CSE/Chem)
William Johnson (CSE)

funded by

ARO NSA NSF